

Decision Tree Master Guide (Telugu Transliteration)

Ee document lo, "DecisionTreeNotes.pdf" lo unna concepts ni base chesukuni, oka sari chadivithe mind lo pratyakshamga gurthundelaga **Decision Trees** gurinchi pin-to-pin Telugu transliteration lo (English letters tho) chala detailed ga explain chestanu. Deentlo intuitive analogies, math calculations, step-by-step coding, mariyu visualizations ni pratyekamga include chesanu.

1. Decision Tree Ante Enti? (The Core Analogy)

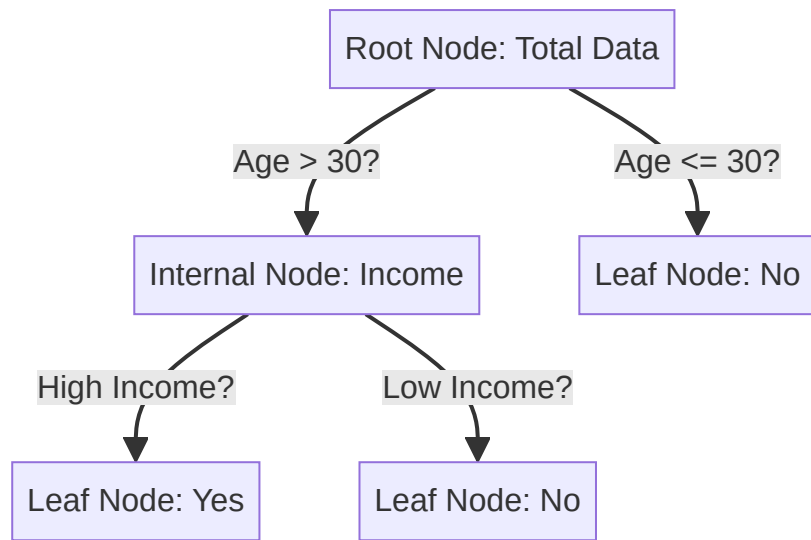
Decision Tree ante manam nija jeevitham lo chese alochana vidhanam (flowchart-like structure) lanti oka Supervised Machine Learning algorithm. Deenini **Classification** mariyu **Regression** rendu rakala problems kosam vadataru.

Mind-Map Analogy: 20 Questions Game

Oka person manasu lo oka animal oohinchukuntadu. Manam "Adi water lo untunda?", "Daaniki wings unnaaya?" ani questions adigi, last ki answer kanukkuntam kada? **Decision Tree kuda exact ga ilage pani chestundi!** Idi data ni "if-else" questions dwara small subsets ga split chestu potundi.

Tree Structure (Mukhya Bhagalu):

- Root Node:** Moktham dataset start ayye top node (first split ikkade jarugutundi).
- Decision / Internal Nodes:** Okariki oka condition pettina nodes (e.g., "Age > 30?").
- Branches:** Condition true leda false ayinappudu velle darulu (Yes / No).
- Leaf / Terminal Nodes:** Final output leda prediction dorike bottom nodes (e.g., "Buy Bike: Yes").



2. Decision Tree Structure Visualization

Sklearn use chesi Iris dataset meeda train chesina Decision Tree structure kinda chudandi:

Decision Tree Structure (Iris Dataset)



3. Core Intuition: Splitting & Impurity (Purity Ela Kolavali?)

Prathi node daggara tree oka mukhya question adugutundi: “**Ye feature and ye threshold point valla data chala pure ga (clean ga) divide avtundi?**”

Ikkada **Purity** (shubhrata / disorder) ni kolavadaniki 3 mukhya measures unnayi:

A. Entropy (Disorder / Ashanthi)

Entropy ante data lo unna ashanti leda uncertainty. $\text{Entropy}(\mathbf{S}) = - \sum_{i=1}^c \mathbf{p}_i \log_2(\mathbf{p}_i)$

- **Entropy = 0**: Node 100% pure (andaru okate class ki chendaru).
- **Entropy = 1**: Maximum impurity (50% Yes, 50% No - chala confusion undi).

B. Information Gain (Entropy Reduction)

Split ki mundu unna entropy ki, split tarwata vachchina weighted child entropy ki unna difference ne **Information Gain** antaru. Tree eppudu **maximum Information Gain** iche feature ni root node ga choose chestundi.

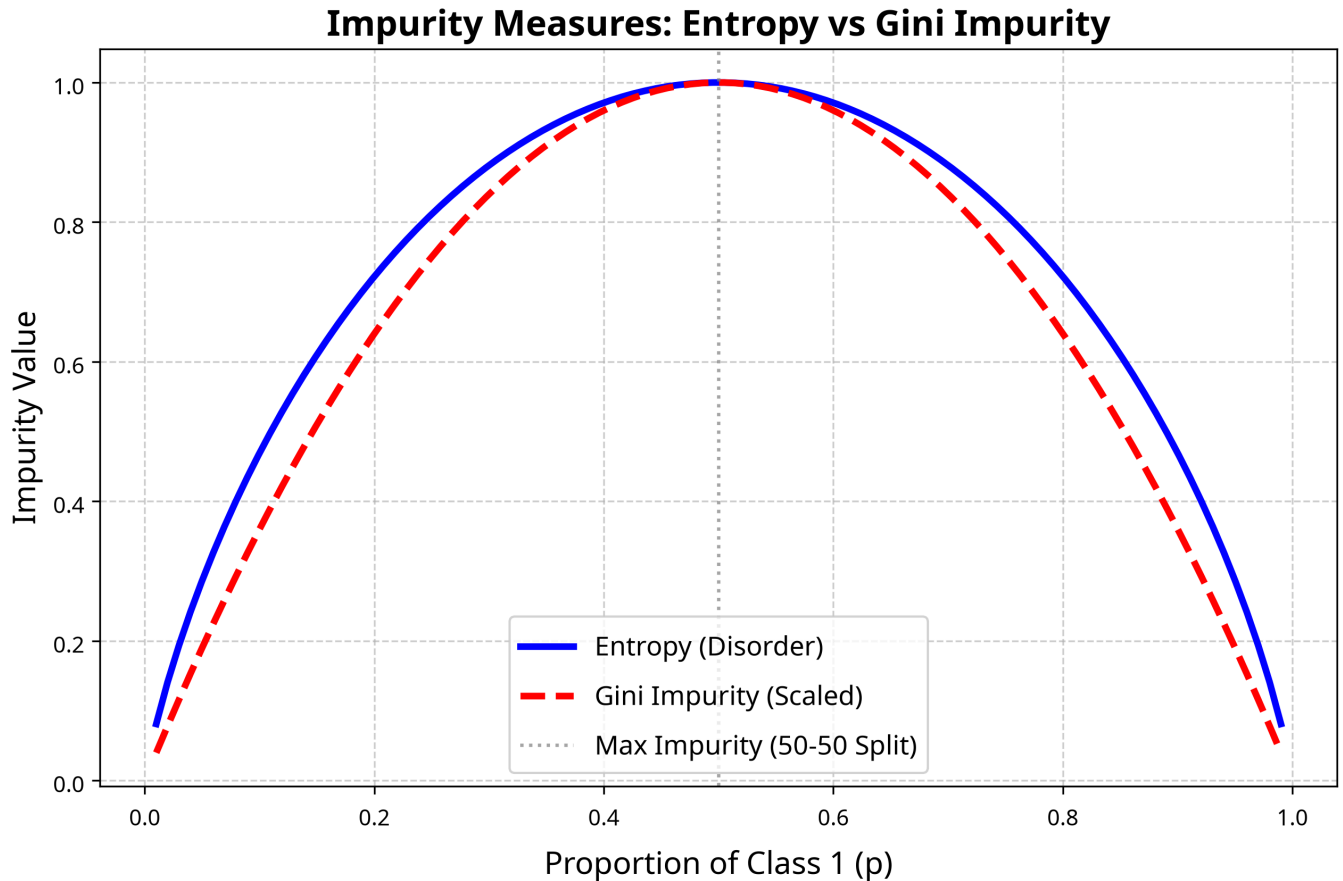
C. Gini Impurity (CART Algorithm Choice)

Classification lo scikit-learn default ga Gini Impurity ni vadutundi. Idi compute cheyadaniki chala fast ga untundi (logarithms undavu). $\text{Gini}(\mathbf{S}) = 1 - \sum_{i=1}^c \mathbf{p}_i^2$

- **Gini = 0**: Pure node.
- **Gini = 0.5**: Maximum impurity (Binary 50-50 split).

Visualization: Entropy vs Gini Impurity Comparison

Entropy mariyu Gini impurity proportion (p) tho ela marutayo kinda curve lo chudandi:



4. Regression Trees (Numbers Predict Cheyadam)

Classification lo classes (Yes/No) predict chesthe, Regression lo continuous numbers (e.g., House Price) predict chestaru.

- **Splitting Criterion:** Entropy/Gini badulu ikkada **Variance Reduction (or MSE - Mean Squared Error)** vadataru.
- **Leaf Prediction:** Leaf node ki vachchina data points yokka **mean (average) value** ne final prediction ga isthundi.

5. MASTER CODING SECTION: Step-by-Step Python Implementation

Meeru error lekunda code rayadaniki, scikit-learn use chesi Decision Tree classifier train chesi, tree plot chuse complete Python code ikkada ichchanu.

```

# =====
# STEP 1: Import Required Libraries
# =====
import numpy as np
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import matplotlib.pyplot as plt

# =====
# STEP 2: Load Dataset
# =====
iris = load_iris()
X = iris.data
y = iris.target

# =====
# STEP 3: Train-Test Split
# =====
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# =====
# STEP 4: Train Decision Tree Classifier
# =====
# max_depth ni limit cheste overfitting tagguthundhi
dt_model = DecisionTreeClassifier(criterion='gini', max_depth=3, random_state=42)
dt_model.fit(X_train, y_train)

# =====
# STEP 5: Predictions & Evaluation
# =====
y_pred = dt_model.predict(X_test)

print("--- DECISION TREE EVALUATION ---")
print(f"Accuracy: {accuracy_score(y_test, y_pred):.4f}")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# =====

```

```
# STEP 6: Feature Importances
# =====
print("\n--- FEATURE IMPORTANCES ---")
for name, imp in zip(iris.feature_names, dt_model.feature_importances_):
    print(f"{name}: {imp:.4f}")
```

6. Overfitting & Pruning (Tree Chala Peddagaithe Em Jarugutundi?)

Overfitting Problem

Decision tree ki depth limit pettakunda vadileste, adi training data lo unna prathi point ni memorise chesi, 100% training accuracy istundi. Kani test data ki vachesariki fail avtundi (**High Variance / Overfitting**).

Pruning (Tree ni Trim Cheyadam)

Overfitting ni agginchadaniki rendu methods unnayi:

1. **Pre-Pruning (Early Stopping)**: Tree perugutunnappude `max_depth`, `min_samples_split`, `min_samples_leaf` lanti restrictions petti aapestham.
2. **Post-Pruning (Cost Complexity Pruning)**: Tree ni purthi ga penchi, tarwata use leni branches ni cut chestam.

7. Real-Life & Data Science Applications

- **Loan Approval Systems**: Bank lo customer salary, credit score, age batti loan approve cheyacha leda ani decide cheyadam.
- **Medical Diagnosis**: Symptoms batti patient ki disease unda leda ani diagnose cheyadam.
- **Customer Churn Prediction**: Customer app unchuntara leda uninstall chestara ani predict cheyadam.

AI/ML Model Recommendations

Single Decision Trees chala sensitive ga untayi (high variance). Anduke data science lo single tree badulu **Ensemble Models** aina **Random Forest**, **Gradient Boosting (GBM)**, **XGBoost**,

LightGBM, mariyu CatBoost ni standard ga vadataru. Ivi chala trees ni combine chesi robust predictions istayi.

References:

1. [DecisionTreeNotes.pdf](#) - Original PDF document for Decision Tree classification, regression, entropy, Gini impurity, and pruning.